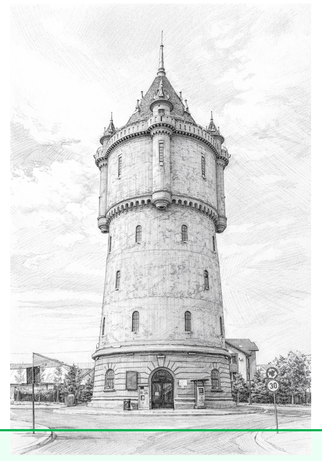
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Transnational Action Plan to boost education on sustainability

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Abstract

The Trans-National Action Plan (TNAP) developed within the ANGIE project (Academic Network for a Green and Innovative Europe) is a strategic framework designed to support the integration of sustainability competencies into European higher education systems. Its necessity arises from the growing challenges generated by climate change, the European Green Deal, digital transformation, and the increasing demand for green skills in the labour market. Universities can no longer operate through isolated national approaches; instead, coordinated transnational cooperation is required to prepare future professionals capable of supporting Europe's green and digital transition.

The TNAP brings together six National Action Plans from Germany, Slovenia, Italy, Portugal, Romania, and Spain into a unified European framework while respecting national educational systems and accreditation regulations. The document aligns with major European strategies such as the European Green Deal, Fit for 55, the European Education Area, and the UN Sustainable Development Goals.

The main content of the TNAP is structured around five strategic pillars. The first pillar focuses on curriculum reform through the integration of sustainability competencies, interdisciplinary learning, and micro-credentials. The second pillar addresses faculty development by promoting green pedagogies, digital teaching methods, and international academic cooperation. The third pillar concentrates on institutional capacity building, including digitalization, green campuses, and sustainability-oriented research infrastructure. The fourth pillar strengthens partnerships between universities, industry, and society through internships, advisory councils, and collaborative innovation initiatives. Finally, the fifth pillar establishes monitoring and evaluation mechanisms based on indicators, surveys, feedback systems, and adaptive governance. The TNAP emphasizes cross-border collaboration through shared educational resources, student and staff mobility schemes, joint research activities, common quality assurance frameworks, and coordinated implementation mechanisms designed to ensure long-term institutional resilience, knowledge transfer, and sustainable impact across the European higher education landscape.

The TNAP also proposes common mobility programmes, shared educational resources, joint research activities, accreditation pathways, risk management strategies, and implementation timelines for 2026–2030. Overall, the document serves both as a policy reference and as an operational roadmap for creating a resilient, sustainable, and innovation-oriented European higher education network.

Key Principles

1. Subsidiarity / EU-level coordination with national implementation flexibility

The principle of subsidiarity underpins the entire TNAP framework. It recognizes that while strategic coordination at the European level is essential for achieving scale and coherence, the actual implementation of actions must remain anchored in national and institutional contexts. This means that consortium-wide priorities are translated into locally relevant measures, taking into account specific regulatory requirements, accreditation procedures, and labour market conditions.

2. Co-creation / Joint development of resources, curricula, and assessment tools

Co-creation is central to the collaborative ethos of the ANGIE project. Rather than developing solutions in isolation, partner institutions work together to design curricula, learning materials, micro-credentials, and assessment frameworks. This collaborative process not only enhances the quality and relevance of outputs but also fosters mutual learning and capacity building across the consortium.

3. Evidence-based / Continuous monitoring using shared indicators and survey instruments

The TNAP adopts an evidence-based approach to planning and implementation, grounded in systematic data collection and analysis. Shared indicators, survey instruments (such as QS2E and QS2S), and monitoring platforms are used to track progress, assess impact, and identify areas for improvement. This continuous feedback loop supports adaptive management, allowing partners to refine strategies in response to emerging trends, policy changes, and stakeholder needs.

4. Inclusivity / Attention to SMEs, regional disparities, and underrepresented groups

Inclusivity is a cross-cutting priority that ensures the benefits of the green transition are widely distributed and socially equitable. The TNAP explicitly addresses the needs of small and medium-sized enterprises (SMEs), which play a crucial role in European economies but often face barriers in accessing skills development and innovation support. It also considers regional disparities, recognizing that different areas may have varying levels of capacity, infrastructure, and access to resources.

Timeline and milestones

Phase 1: Foundation and Alignment (2026)

The initial phase focuses on establishing the operational and strategic foundations necessary for effective implementation. Key milestones include the formation of national coordination structures, the alignment of institutional strategies with TNAP priorities, and the finalization of core shared instruments such as the ANGIE Competency Framework and the OER repository.

Phase 2: Implementation and Scaling (2027–2028)

The second phase represents the core implementation period, during which pilot actions are expanded and integrated into institutional practices. Interdisciplinary modules, micro-credentials, and joint learning activities are rolled out at scale, while faculty training programmes and university–industry partnerships are actively developed. "Living lab" initiatives and digital learning tools are operationalized, connecting education with real-world WATERIC challenges.

Phase 3: Consolidation and Sustainability (2029–2030)

The final phase focuses on consolidating results, institutionalizing successful practices, and ensuring long-term sustainability beyond the project lifecycle. Key milestones include the integration of validated approaches into national and institutional strategies, the publication of comparative progress reports, and the execution of external evaluations.

Risk management

Risk	Probability	Impact	Mitigation Strategy
Fragmented implementation across national systems	High	High	• Maintain strong Steering Committee coordination • Use "minimum viable harmonization" approach • Celebrate and share early wins to build momentum
Funding volatility post-2027 EU programming period	Medium	High	• Diversify funding sources early • Embed TNAP priorities in institutional strategic plans • Advocate for continued EU support via CRUE, EUA, etc.
Accreditation delays for joint/interdisciplinary programs	Medium	Medium	• Engage QA agencies early in design phase • Pilot modules before full program submission • Document evidence of labour market relevance
Uneven capacity among partners (resources, expertise)	High	Medium	• Implement peer-support pairings • Allocate consortium funds for capacity-building • Rotate leadership roles to distribute experience
Political/regulatory shifts in member states	Medium	Medium	• Maintain non-partisan, evidence-based positioning • Build coalitions with business/NGO stakeholders • Keep implementation flexible to adapt to policy changes

Monitoring and evaluation framework

Core indicator set (2026 Baseline → 2030 Targets)

Dimension	Indicator	Baseline (est.)	2028 Target	2030 Target
Curriculum	% of degree programs with embedded green learning outcomes	15%	35%	65%
Faculty	# of educators completing ANGIE CPD in sustainability pedagogy	50	300	1,200
Students	% of students completing internships/projects in WATERIC sectors	8%	25%	50%
Partnerships	# of formalized industry/university collaborations under TNAP	20	60	150
Resources	# of OERs/case studies contributed to consortium repository	30	150	500
Impact	Graduate employability in green roles (6 months post-graduation)	TBD	+15% vs. baseline	+30% vs. baseline

Shared strategic pillars

Pillar 1: Transversal curriculum reform

Objective	Consortium-wide actions	National adaptations
Embed sustainability competencies across all disciplines	• Develop a common ANGIE Competency Framework aligned with GreenComp and EQF Level 7 • Create modular, stackable micro-credentials in WATERIC domains • Establish a shared repository of Open Educational Resources (OERs) in multiple languages	• Germany: Coordinate via KMK respecting Länder autonomy • Italy/Romania: Align with ANVUR/ARACIS accreditation cycles • Slovenia/Portugal/Spain: Leverage existing national greening strategies
Promote interdisciplinary, project-based learning	• Launch a Consortium Case Study Bank with real-world WATERIC challenges from each country • Develop joint capstone projects connecting students across borders via virtual collaboration	• Adapt case studies to local regulatory/industrial contexts • Ensure language accessibility for cross-border student teams

3.2 Pillar 2: Faculty capacity building & pedagogical innovation

Objective	Consortium-wide actions	National adaptations
Equip educators with green content knowledge and active-learning pedagogies	• Establish an ANGIE Trainer Network with "train-the-trainer" programs • Offer blended CPD modules (1-3 months) on eco-pedagogy, digital tools, and interdisciplinary teaching • Create a peer-mentoring platform for cross-institutional exchange	• Schedule training to align with national academic calendars • Provide materials in national languages while maintaining English versions for mobility
Incentivize teaching innovation	• Advocate for inclusion of sustainability teaching excellence in promotion criteria across consortium institutions • Launch an annual ANGIE Teaching Innovation Award	• Work within national HR frameworks (e.g., Italy's ASN, Spain's ANECA criteria)

3.3 Pillar 3: Institutional capacity and Digital infrastructure

Objective	Consortium-wide actions	National adaptations
Modernize infrastructure as "Living Labs"	• Develop common guidelines for Green Campus Certification aligned with NZEB standards • Share best practices on energy monitoring, circular procurement, and biodiversity management	• Adapt to national building codes and funding mechanisms (e.g., PNRR Italy, RRP Slovenia)
Accelerate digitalization for sustainability education	• Co-develop simulation tools, digital twins, and AI-assisted learning platforms for WATERIC scenarios • Ensure interoperability with European digital education infrastructure (EduGAIN, Europass)	• Comply with national data protection regulations while enabling cross-border access

3.4 Pillar 4: University-Industry-Society partnerships

Objective	Consortium-wide actions	National adaptations
Align education with labour market needs	• Create a Consortium Industry Advisory Council with representatives from WATERIC sectors in each country • Standardize internship frameworks with learning outcomes and assessment rubrics	• Tailor partnerships to national industrial structures (e.g., SME focus in Portugal/Spain; large industry in Germany/Italy)
Support SMEs in sustainability innovation	• Develop low-threshold engagement formats: short courses, challenge briefs, mentorship schemes • Pool resources for a SME Green Skills Toolkit translated into all consortium languages	• Leverage national business associations (BDI, DIHK, Confindustria, etc.) for outreach

3.5 Pillar 5: Monitoring, evaluation and adaptive governance

Objective	Consortium-wide actions	National adaptations
Track progress with shared indicators	• Adopt a Core Indicator Set (see Section IV) administered via the QS2E/QS2S survey instruments • Publish annual ANGIE Progress Reports with comparative dashboards	• Integrate data collection into national QA systems (ANECA, ARACIS, ANVUR, NAKVIS, etc.)
Enable continuous adaptation	• Hold biannual Consortium Review Forums to share lessons and adjust strategies • Establish rapid-response working groups for emerging challenges (e.g., new EU regulations)	• Ensure feedback loops inform national NAP revisions